MAIS PAC

IPhO 04.01: Algebra, Functions and Geometry · Lecture + Problems

Algebra, Functions and Geometry

MAIS Physics & Astronomy Club (PAC) · IPhO lecture + problem handout

6 lecture sections · 10 problems · 1 interactive experiment · 5 vector figures

Start with a physical picture, then introduce only the mathematics needed to make it predictive. The lecture connects everyday reasoning to olympiad measurements. Five vector diagrams and an interactive experiment support the argument; ten graded problems and independent worked solutions turn the ideas into practice. The coverage checklist is the IPhO syllabus, not this handout.

Contents & learning path

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1. The algebra that physics actually uses

Mathematics papers test technique; a physics problem tests whether you can **choose** the technique. The algebra you need is narrow and repetitive: solve a pair of linear equations, complete a square, take a ratio from similar triangles, use one trigonometric identity, and handle a power or root without slipping a factor.

For an IPhO problem the most valuable habit is to do the algebra **symbolically first**. Numbers introduced early hide structure and make a mistake unrecoverable; a symbolic result shows which combination governs the physics.

2. Functions: shapes you should be able to sketch

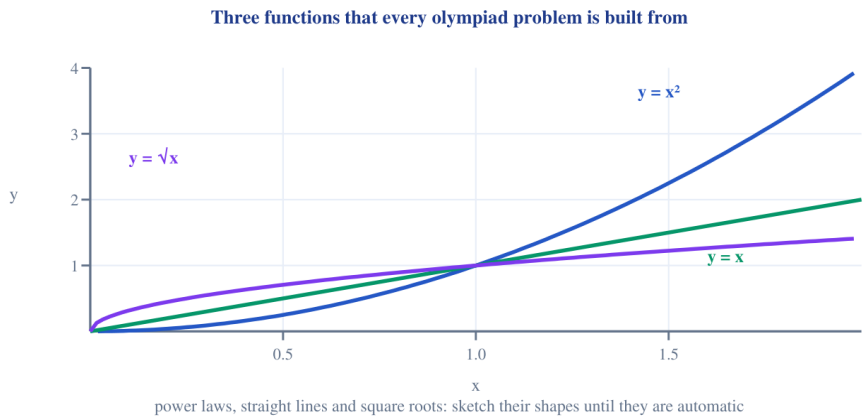


Figure 1. Knowing the shape, the gradient and the growth of the basic functions lets you sketch a solution before you calculate anything — and spot at once when an answer must be wrong.

function	shape	grows like	typical use
x^n, n>0	through the origin, flat then steep	x^n	area, volume, energy
x^(1/2)	steep then flat	slow	period of a pendulum, orbital speed
1/x	hyperbola	falls fast	Coulomb, gravity, pressure
1/x²	falls faster	—	inverse-square fields
sin x,cos x	oscillation	bounded	waves, circular motion
e^(-x)	decay	bounded	cooling, RC circuits

Two algebraic manipulations recur so often that they are worth memorising:

- **Completing the square:** $ax^2+bx+c = a(x+(b)/(2a))^2 + c - (b^2)/(4a)$. This reads off the vertex of a parabola immediately, which is usually the physically interesting point (maximum range, minimum energy, critical damping).
- **Rationalising:** $(1)/(\sqrt{(a)}-\sqrt{(b)}) = (\sqrt{(a)}+\sqrt{(b)})/(a-b)$. This turns a small difference of large quantities — the usual source of catastrophic cancellation in numerical work — into a stable expression.

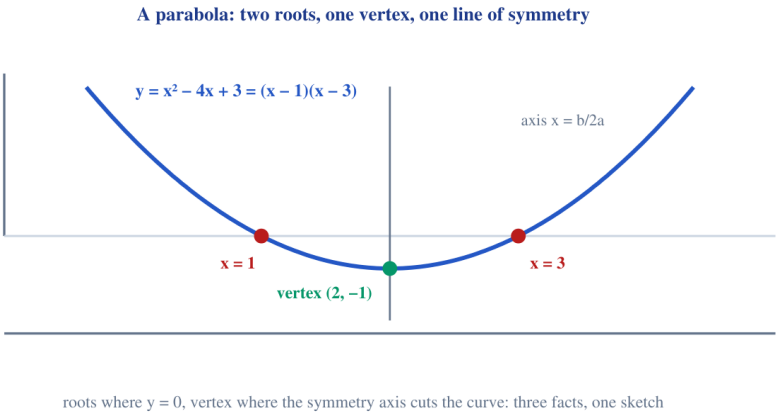


Figure 2. Completing the square, $y = a(x + b/2a)^2 + c – b^2/4a$, puts the vertex and the axis of symmetry on show, and the discriminant $b^2 – 4ac$ counts the real roots.

3. Geometry: similarity before trigonometry

When two angles of a triangle agree, all three sides are in a fixed ratio. Almost every "find the length in this diagram" problem is a similarity problem in disguise, and writing the proportion is faster and less error-prone than setting up a coordinate system.

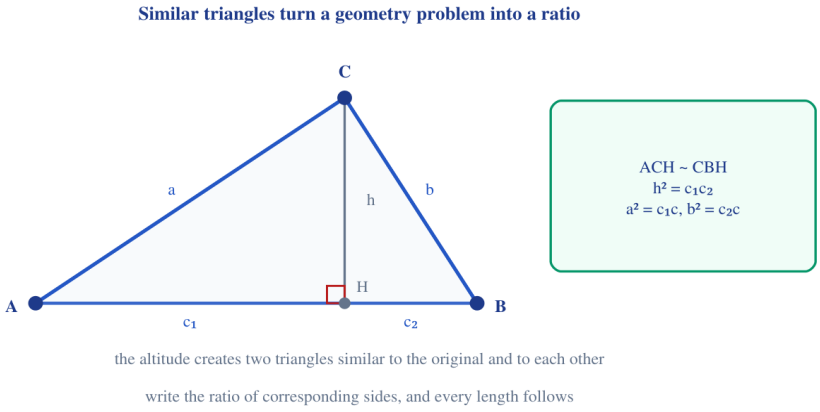


Figure 3. Similarity is the olympiad workhorse: once two angles agree, every length ratio is fixed, and a single proportion replaces pages of trigonometry.

The altitude rule in a right triangle is the standard example: the altitude to the hypotenuse creates two smaller triangles, each similar to the original, so

$$h^2 = c_1c_2,\ a^2 = c_1c,\ b^2 = c_2c. \tag{1}$$

These three are Pythagoras plus similarity, and they solve most ladder-and-shadow problems in one line.

For circles, three facts cover a great deal:

- The angle at the centre is twice the angle at the circumference on the same arc.
- A tangent is perpendicular to the radius at the point of contact.
- The power of a point: for a point P outside a circle and any secant through it, $PT^2 = PA \cdot PB$ where T is the tangent contact point.

4. Trigonometry

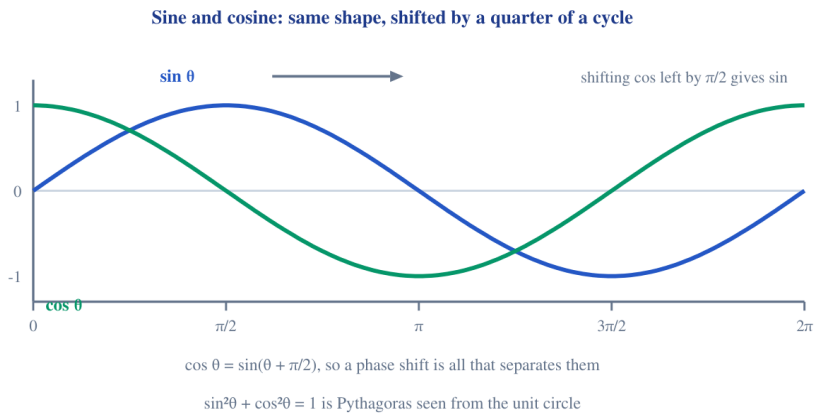


Figure 4. Phase, amplitude and period are the three numbers that describe any harmonic quantity; a graph showing all three is quicker to read than a formula.

Beyond the definitions, three tools do nearly all the olympiad work:

- **The sine rule** (a)/(sin A) = (b)/(sin B) = 2R and the **cosine rule** $c^2 = a^2+b^2-2ab\cos C$. The cosine rule is the generalisation of Pythagoras; whenever a problem has two known sides at a known angle, this is the tool.
- **Small angles:** for θ in radians, sin θ ≈ θ, tan θ ≈ θ, cos θ ≈ 1 - θ²/2. These are the approximations that make a pendulum simple-harmonic.
- **Products to sums:** 2sin Asin B = cos (A-B) - cos (A+B) and its relatives, which convert a beat pattern or a standing wave into two travelling waves.

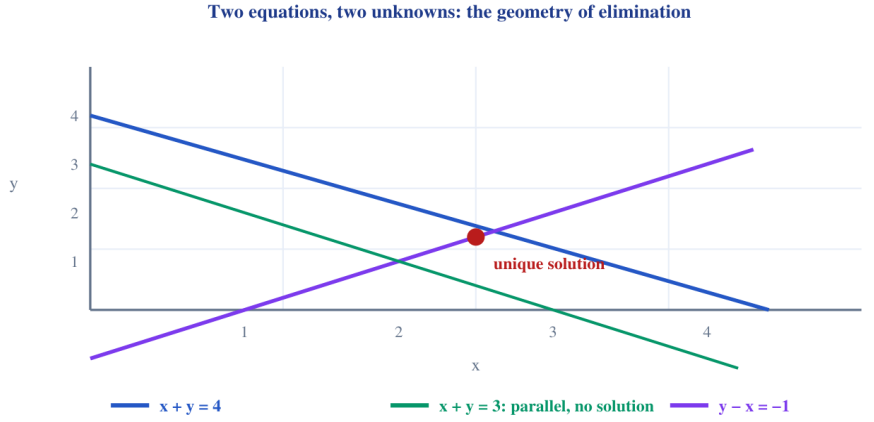


Figure 5. A pair of linear equations is a pair of lines: one intersection, no intersection, or a whole line of them. Elimination is arithmetic; the picture tells you which case you are in.

5. Simultaneous equations and constraints

A physics problem with N unknowns needs N independent statements. The standard procedure:

1. Write the constraint equations (geometry, conservation, equilibrium).
2. Count them; if there are fewer than unknowns, you have missed a condition — usually a boundary condition or a symmetry.
3. Eliminate symbolically, not numerically, so that the final expression shows the physics.
4. Check the dimensions of every intermediate line, not only the last one.

Interactive experiment: The quadratic explorer — open the online handout to use it.

Algebra in a physics answer. A grader can only award method marks for algebra they can follow. So: one relation per line, no arithmetic in your head, symbols until the last line, and a dimensional check at every stage. A symbolic answer that is dimensionally right and reduces correctly in a limiting case is usually worth most of the marks even if a numerical slip spoils the final number.

6. Problems: reason, calculate, check

Ten original problems in three tiers: **[1]** foundation, **[2]** multi-step olympiad reasoning, **[3]** challenge. Use the constants supplied in the lecture unless a question overrides them. State assumptions, keep units, and estimate the answer before computing it. Every problem has a worked solution with a consistency check.

[1] Problem 1 — Completing the square. A projectile launched at speed v at angle θ has range R = (v²sin 2θ)/(g). (a) For fixed v, what angle maximises R? (b) A ball is thrown from ground level and lands 40 {m} away with v = 20 {m,s^(-1)}. Find the two possible launch angles.

Worked solution →

[1] Problem 2 — A ratio from similarity. A tree casts a shadow 12.0 {m} long. At the same moment a vertical pole 1.80 {m} high casts a shadow 2.40 {m} long. (a) Find the height of the tree. (b) Explain why the two triangles are similar. (c) State one physical reason the result could be systematically wrong.

Worked solution →

[2] Problem 3 — Two equations, two unknowns. Two blocks connected over a pulley have masses m_1 = 3.0 {kg} and m_2 = 5.0 {kg}, with friction negligible. (a) Write the two equations of motion. (b) Solve them for the acceleration and the tension. (c) Check the tension in the limit m_1 = m_2.

Worked solution →

[2] Problem 4 — The cosine rule in a field problem. Two charges +q and +q are separated by d. A third charge +q is placed at a point a distance d from each of them, so the three charges sit at the vertices of an equilateral triangle of side d. (a) State the angle between the two forces on the third charge. (b) Find the magnitude of their resultant. (c) Explain why moving the two fixed charges closer together increases the resultant faster than 1/r².

Worked solution →

[2] Problem 5 — Small-angle approximations. A simple pendulum of length L = 1.00 {m} is released from 5.0°. (a) Show that the restoring force is proportional to the displacement for small angles. (b) Compute the fractional error in using sin θ ≈ θ at 5.0°. (c) At what angle does the error reach 1%?

Worked solution →

[2] Problem 6 — A discriminant condition. A ball is thrown from a cliff of height h at speed v. (a) Write the quadratic whose roots give the times of flight to the ground. (b) Show that a real root always exists. (c) Explain physically why the discriminant cannot be negative here, and give a situation in which it can.

Worked solution →

[2] Problem 7 — Inverse-square geometry. The intensity of a point source falls as $1/r^2$. (a) Show that doubling the distance reduces the intensity to one quarter. (b) Two identical lamps are 4.0 {m} apart. Find the point on the line between them where the illuminance is the same from both. (c) Explain why the answer does not depend on the power of the lamps if they are identical.

Worked solution →

[3] Problem 8 — Minimising a function. A block of mass m is pulled along a rough floor by a force at angle θ above the horizontal, with coefficient of friction μ. (a) Write the pulling force needed for constant speed as a function of θ. (b) Show that it is minimised when tan θ = μ. (c) Evaluate the minimum force for m = 10 {kg}, μ = 0.40.

Worked solution →

[3] Problem 9 — Three equations, one result. A uniform rod of length L and mass M rests with one end against a smooth wall and the other on a rough floor, at angle θ to the floor. (a) Write the equilibrium equations. (b) Show that equilibrium requires μ (1)/(2)θ. (c) Interpret the result as θ becomes large.

Worked solution →

[3] Problem 10 — An identity in disguise. Two coherent waves of equal amplitude A and angular frequency ω travel in the same direction with a phase difference φ. (a) Use a trigonometric identity to write the sum as a single wave. (b) Find the amplitude of the sum. (c) At what phase difference is the resultant zero, and what does this mean physically?

Worked solution →

Further practice and provenance

The 10 exercises above and their solutions are **original MAIS PAC problems**, inspired by the skills in this IPhO syllabus topic. They are not copied from official papers and are not official marking schemes.

Sources and scope. Topic coverage follows the IPhO syllabus (accepted 2014, amended 2015) as published by the IPhO organisation. Official problems are available through the IPhO past-paper archive. The derivations, diagrams, experiments and numerical teaching examples in this handout were prepared independently for MAIS PAC. Calculator-level methods are emphasized; optional calculus connections are explicitly identified. Synthetic data, when used, are labelled as teaching data.

Where to learn more

IPhO syllabus and statutes

The coverage checklist this curriculum follows: accepted 2014, amended 2015. Read it once for scope, not for depth.

IPhO past problems and solutions

The only honest test of readiness. Work recent theoretical and experimental papers under time, then compare with the official marking schemes.

Paul's Online Math Notes

Calculus, series and approximation practice at exactly the level the syllabus expects.